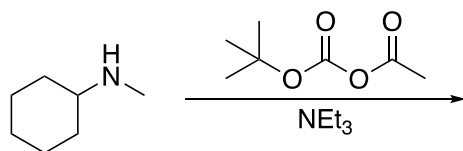


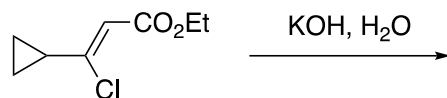
Acylation of Nucleophiles; Carboxylic Acids

1. Provide the neutral organic product of each reaction below after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product.

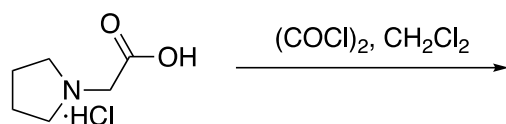
A.



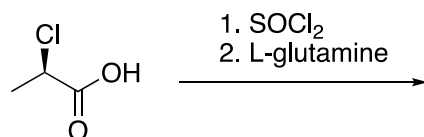
B.



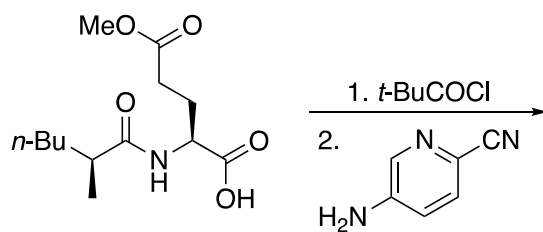
C.



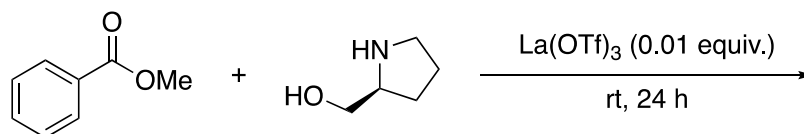
D.



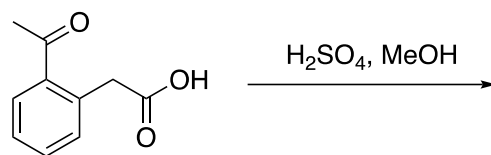
E.



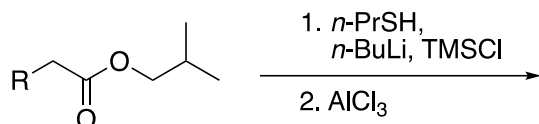
F.



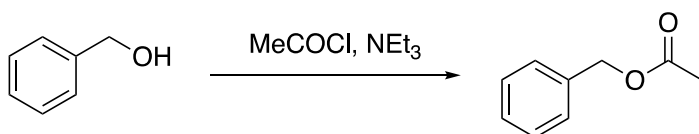
G.



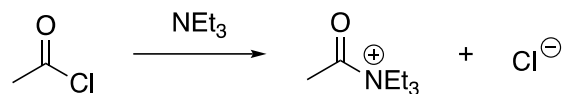
H.



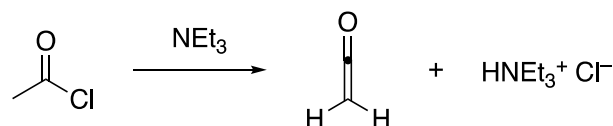
2. Alcohols react with acyl chlorides in the presence of tertiary amines to yield esters. There are two plausible mechanisms for this reaction, both of which begin with reaction between the acyl chloride and amine. Mechanism **A** includes an acylammonium intermediate and mechanism **B** a ketene intermediate.¹



- A. Draw a curved-arrow mechanism for formation of the acylammonium intermediate in mechanism **A**.



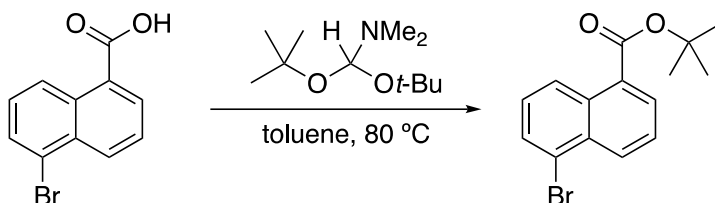
- B. Draw a curved-arrow mechanism for formation of the ketene intermediate in mechanism **B**.



- C. To distinguish between these mechanisms, the cited authors used deuterated methanol (MeOD) as the nucleophile. Different ester products would result upon the operation of mechanism **A** or **B**. Provide both products with their associated mechanisms.
- D. When NEt_3 is used as in the example above, a mixture of the two products results. Use of pyridine ($\text{C}_5\text{H}_5\text{N}$) results in the almost exclusive operation of mechanism **A**. Propose an explanation for the difference in mechanism between the two amines.

¹ Truce, W. E.; Bailey Jr., P. S. *J. Org. Chem.* **1969**, 34, 1341. ([link](#))

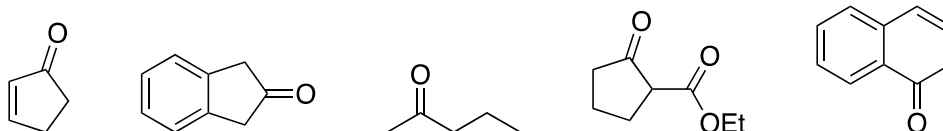
3. The di-*tert*-butyl acetal of dimethylformamide (DMF) is effective for alkylation of carboxylic acids in what amounts to a nucleophilic substitution reaction at the acetal carbon.



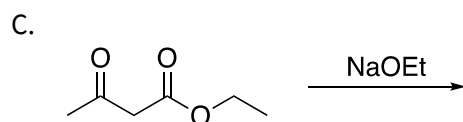
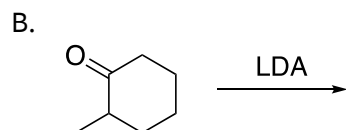
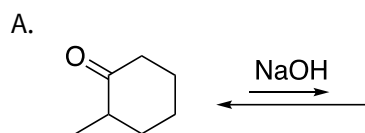
- To begin drawing a mechanism for this reaction, find the most acidic group and the most basic group in the starting materials and draw the products of proton transfer between these groups.
- The conjugate acid thus produced can fall apart to produce the *tert*-butyl cation and two other products. Use curved arrows to depict electron flow for this E2 elementary step and draw the two other products. Then, complete the mechanism by drawing curved arrows for an A_N step to the *tert*-butyl cation.

Enols and Enolates; α -Functionalizations of Carbonyl Compounds

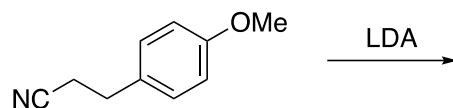
4. Draw the dominant enol tautomer of each compound in solution and rank the compounds from largest to smallest amount of enol in solution at equilibrium.



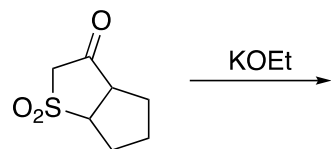
5. Provide the structure of the conjugate base of each organic substrate produced by the action of the indicated base.



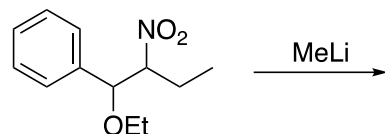
D.



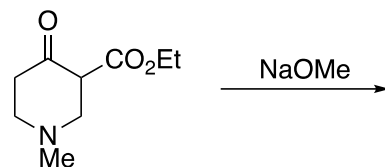
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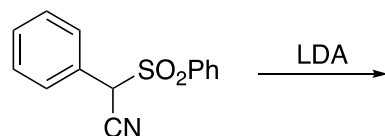
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G.

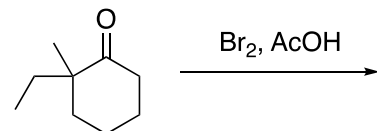


H.

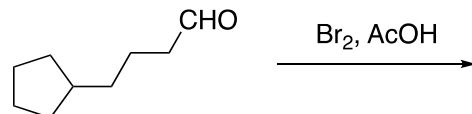


6. Provide the neutral organic product of each reaction below after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product.

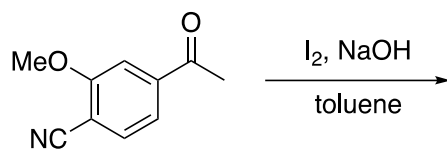
A.



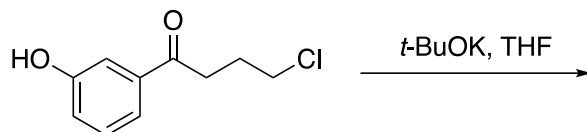
B.



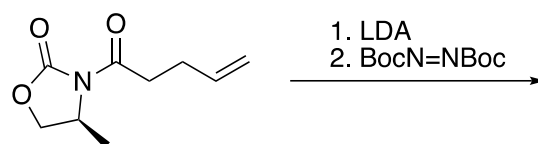
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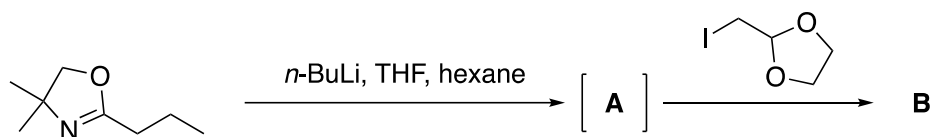
D.



E.

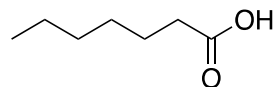


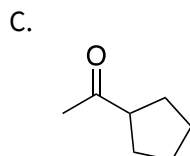
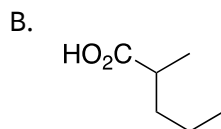
7. Oxazolines like the one below have been used as carboxylic acid equivalents. They can be alkylated and subsequently converted to carboxylic acids, revealing α -substituted products that can't be made via direct alkylation.



- A. Determine the most acidic position of the organic substrate above and draw the conjugate base of the substrate formed upon treatment with n -butyllithium (**A**). Then, draw the product **B** of reaction of the anion **A** with the alkyl iodide shown.
- B. Treatment of **B** with methyl iodide followed by 1 M NaOH(aq) converts the oxazoline to a carboxylic acid. Propose a mechanism for this reaction. List the elementary steps of your mechanism in your response but do not provide curved arrows.
8. Starting from compounds containing five carbons or less and using inorganic reagents of your choice, devise syntheses of the following compounds.

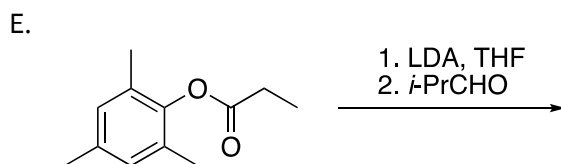
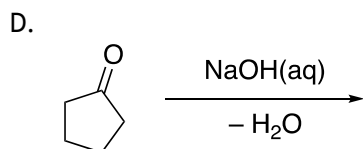
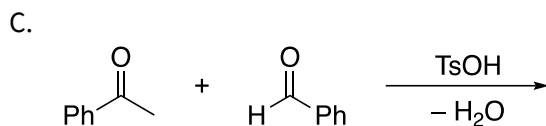
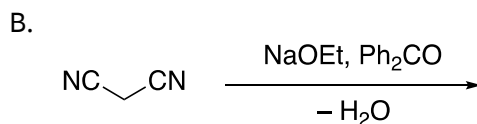
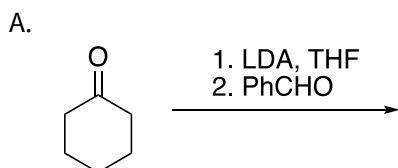
A.



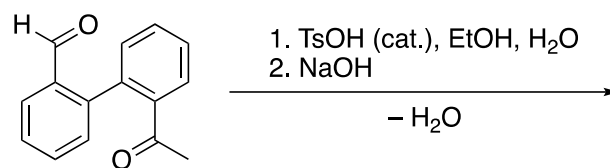


Aldol and Aldol-esque Reactions: Additions and Condensations

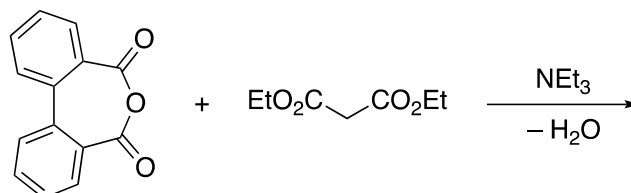
9. Provide the neutral organic product of each reaction below after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product. “– X” indicates loss of a small molecule X in a condensation process.



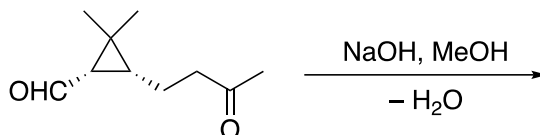
F.



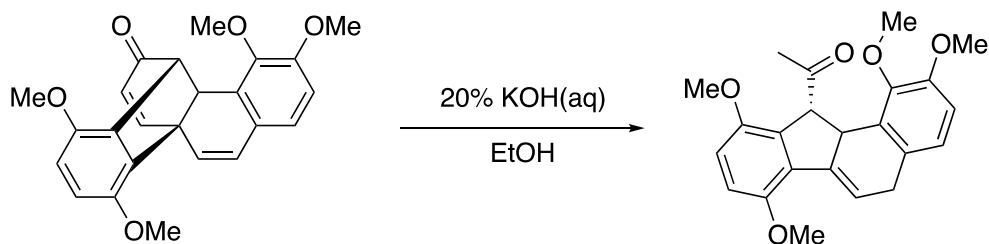
G.



H.



10. The reaction below combines a retro-aldol process with addition-elimination at a carbonyl group. The substrate reacts with two equivalents of hydroxide and the only byproduct is potassium formate.



- Draw the product of retro-aldol condensation of the starting material. The structure should include two carbonyl groups.
- Use curved arrows to show how addition-elimination followed by proton transfer leads to the product shown as well as formate anion. The sequence of elementary steps is 1. Ad_N; 2. E_β; 3. pt.